

Q1.

A supermarket buys pears from a local supplier. The supermarket requires the mean weight of the pears to be at least 175 grams. William, the fresh-produce manager at the supermarket, suspects that the latest batch of pears delivered does not meet this requirement.

- (a) William weighs a random sample of 6 pears, obtaining the following weights, in grams.

160.6 155.4 181.3 176.2 162.3 172.8

Previous batches of pears have had weights that could be modelled by a normal distribution with standard deviation 9.4 grams. Assuming that this still applies, show that a hypothesis test at the 5% level of significance supports William's suspicion.

(7)

- (b) William then weighs a random sample of 20 pears. The mean of this sample is 169.4 grams and $s = 11.2$ grams, where s^2 is an unbiased estimate of the population variance.

Assuming that the population from which this sample is taken has a normal distribution but with unknown standard deviation, test William's suspicion at the 1% level of significance.

(5)

- (c) Give a reason why the probability of a Type I error occurring was smaller when conducting the test in part (b) than when conducting the test in part (a).

(1)

(Total 13 marks)

Q2.

The marks, X , obtained by students in their mathematics examination may be assumed to be a random variable with mean μ and variance 32.

After the introduction of a new programme of revision classes, it is claimed that the value of the mean has increased from its previous value of 65.

In order to investigate this claim, a random sample of 50 students' marks is taken and the mean, $\bar{x}$, calculated.

- (a) Write down null and alternative hypotheses in order to investigate the claim.

(2)

- (b) Assuming H_0 to be true, write down the distribution of $\bar{X}$.

(2)

- (c) It is decided to reject the null hypothesis if the sample mean exceeds 66.4.

Calculate the probability of a Type I error and hence write down the significance level of this test.

(4)

- (d) Calculate the probability of a Type II error in the case when μ has increased to 67.

(4)

(Total 12 marks)

Q3.

A biased six-sided die is to be rolled once. The probabilities of scoring 1, 2, 3 and 4 are each equal to $\frac{1}{8}$. The probability of scoring 5 is equal to the probability of scoring 6. The score is represented by a random variable X .

- (a) Copy and complete the following table to show the distribution fully.

x	1	2	3	4	5	6
$P(X = x)$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$		

(1)

- (b) Calculate the mean and variance of X .

(5)

(Total 6 marks)

Q4.

The volume, Y millilitres, of washing-up liquid contained in a plastic bottle is a normal random variable with mean μ and standard deviation 8.

It is suspected that μ has a value which is less than the specified value of 500.

To test this suspicion, a random sample of 100 bottles of washing-up liquid is selected and the mean volume, $\bar{Y}$, of liquid per bottle is calculated.

- (a) Write down null and alternative hypotheses for this test.

(1)

- (b) State the distribution of $\bar{Y}$, assuming the null hypothesis to be true.

(2)

- (c) Assuming a 5% significance level:

- (i) write down the probability of a Type I error;

(1)

- (ii) determine the critical region for $\bar{Y}$, giving its limit to two decimal places.

(3)

- (d) Hence determine the probability of a Type II error when $\mu = 497.5$.

(7)

(Total 14 marks)

Q5.

A random variable X is normally distributed with mean μ and standard deviation 0.8. The null hypothesis $H_0 : \mu = 40$ is to be tested against the alternative hypothesis $H_1 : \mu \neq 40$ using the 5% level of significance.

The mean, $\bar{X}$, of a random sample of 50 observations of X is to be used as the test statistic.

- (a) Write down:
- (i) the distribution of X assuming H_0 is true; (3)
 - (ii) the probability of a Type I error. (1)
- (b) Calculate the acceptance region for $\bar{X}$, giving its limits to two decimal places. (5)
- (c) (i) Explain what is meant by a Type II error. (1)
- (ii) Calculate the probability of a Type II error in the case when $\mu = 40.3$. (5)
- (Total 15 marks)**

Q6.

Santos, a newsagent, employs schoolchildren to deliver newspapers. Marilyn is to start delivering morning newspapers. She wishes to complete her delivery round early enough to ensure that she will not be late for school. Santos tells her that the average time taken to complete a delivery round is 40 minutes. Marilyn's friend Lula, who already delivers morning papers for Santos, says that the average time taken to complete a delivery round is 50 minutes.

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Before starting her employment, Marilyn checks the times taken by 8 schoolchildren, employed by Santos, to complete their rounds. The times, in minutes, may be regarded as a random sample from a normal distribution and were as follows:

52 30 41 49 61 33 52 54

- (a) (i) By carrying out a hypothesis test, using the 5% significance level, show that the null hypothesis that the mean time to complete a round is equal to 40 minutes is accepted.
- (ii) By carrying out a hypothesis test, using the 5% significance level, show that the null hypothesis that the mean time to complete a round is equal to 50 minutes is accepted. (12)
- (b) The following three claims refer to the two tests carried out in part (a).
- Claim 1** A Type I error has occurred in at least one of the tests.
- Claim 2** A Type II error has occurred in both tests.

Claim 3

A Type II error has occurred in at least one of the tests.

For **each** of the three claims, state whether it is:

- A** definitely true;
- B** possibly true;
- C** definitely not true.

Explain your answers.

(6)

(Total 18 marks)

Q7.

A make of car battery is advertised as having a mean lifetime of at least 40 months. A car magazine suspects that the mean lifetime is less than 40 months and so monitors the lifetimes, in months, of a random sample of 12 batteries, with the following results:

44 32 46 39 38 30 35 40 38 41 34 36

- (a) Assuming that lifetimes are normally distributed, investigate the magazine's suspicion, using the 5% significance level.

(10)

- (b) A larger random sample of 160 car batteries is now monitored. The lifetimes are found to have a mean of 39.2 months and a standard deviation of 4.2 months. Use this information to investigate the magazine's suspicion, using the 5% significance level.

(4)

- (c) State, with an explanation, which, if either, of the tests in parts (a) and (b) has the larger risk of:

- (i) a Type I error;
- (ii) a Type II error if the mean lifetime is 40 months;
- (iii) a Type II error if the mean lifetime is 39 months.

(6)

(Total 20 marks)

Q8.

Priscilla supplies chilled food cabinets for use when displaying perishable foods in shops. She claims that the mean time to spoilage of yoghurt displayed in one of her cabinets is more than 200 hours.

To test this, Rio, a shopkeeper, placed 11 randomly selected tubs of fresh yoghurt in one of Priscilla's cabinets with a 'Not for Sale' sign. The times to spoilage, in hours, of the yoghurt in these tubs were as follows.

206 213 197 215 202 200 209 217 195 203 204

- (a) Test Priscilla's claim using the 5% significance level. You may assume that the data is a sample from a normal distribution with standard deviation 6.0.

(7)

- (b) Discuss the problems, if any, which would have arisen in the interpretation of the results of the test if three of the tubs of yoghurt had accidentally been sold and so data were only available for the remaining eight tubs. (2)
- (c) Define a Type I error in the context of this question. (2)
- (d) Adam, another shopkeeper, intends to repeat Rio's data collection and analysis. State the probability of a Type I error occurring in the analysis of Adam's data when the mean time to spoilage of the tubs of fresh yoghurt displayed in Priscilla's cabinets is:
- (i) 200 hours;
- (ii) 202 hours.

(3)

(Total 14 marks)



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